

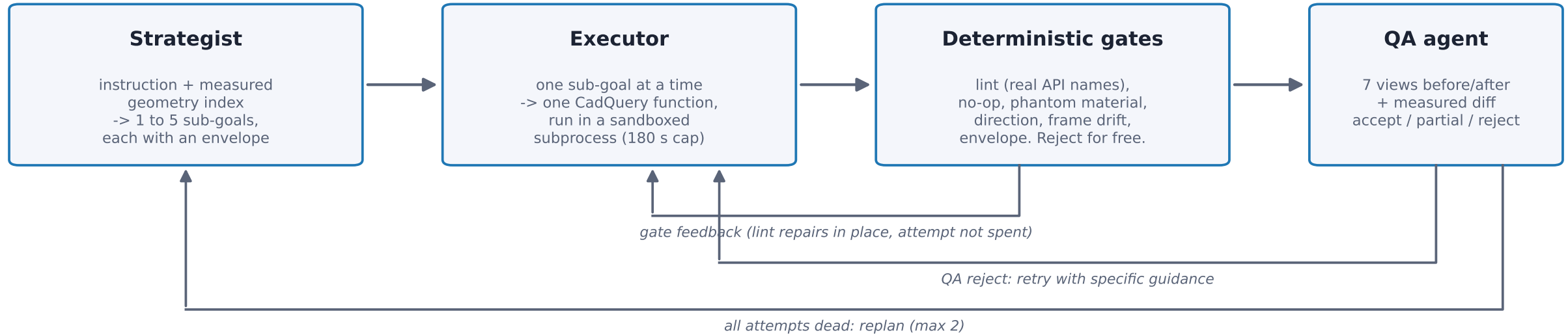
UCONN CAD PACK

Text-instructed 3D CAD editing with a three-agent CadQuery harness

neuralCAD-Edit benchmark | ASME IDETC-CIE 2026 Hackathon (Autodesk)

48 of 48 edit tasks completed | Diff F1 0.37 vs 0.18 for the best published model baseline

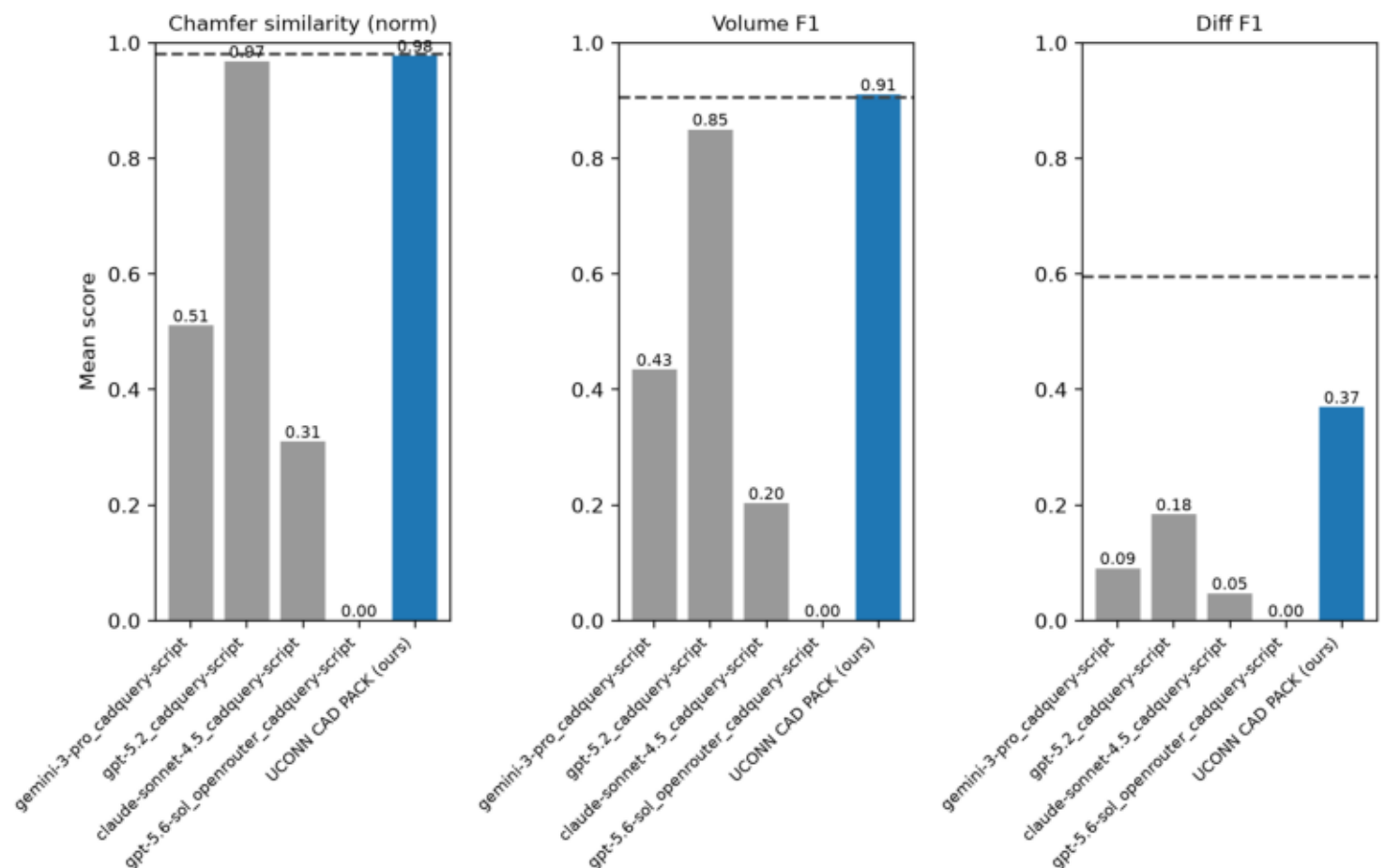
Method: plan, execute, gate, verify



- A stateful router (ADK style) owns the loop: it re-indexes geometry after each accepted sub-goal and forwards QA feedback to the next attempt.
- The strategist sees a measured index, not a picture: hole families by radius, blind vs through openings, paired bores, face areas. Colour renders are attached only when the sentence carries an appearance, view or dimension word.
- Six deterministic gates reject bad geometry before any model judges it. The lint gate asks the installed CadQuery and OCP whether every attribute exists and repairs the name in place, so a typo never burns an attempt.
- QA is a separate model call with no stake in the edit passing. It sees the whole plan, so it never rejects a step for not doing the next step's work.
- Everything is measured against the geometry each attempt started from: volume delta, new-face regions, bounding box drift. Renders illustrate, measurements decide.

Final scores, benchmark metrics on all 48 edit tasks

Metric comparison across models (edit)



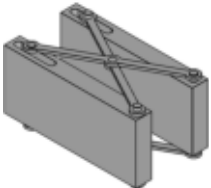
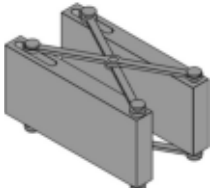
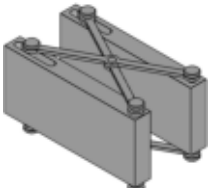
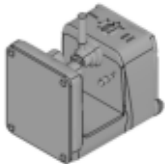
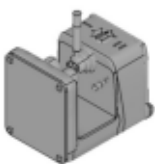
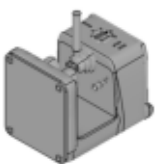
Means over the full split

Surface Chamfer similarity	0.978
Volumetric F1	0.910
Volumetric Difference F1	0.369
Tasks completed	48 / 48
Mean cost per edit	\$0.86

Scores computed with the benchmark's own metric code (src/utils/evals_diff.py and evals_feature_geometric.py), same defaults as the published baselines. Diff F1, the most important metric, is 2x the best published model baseline (0.18); the human reference line is 0.60.

The five selected test examples

Input part, our edit, and the human expert's edit (ground truth), toprightiso view. Scores are chamfer / volume F1 / diff F1.

	Input	Ours	Ground truth	
[medium]				
Please convert the round edges of the gear into straight spur gear teeth.				0.991 0.972 0.131
[medium]				
Add cylindrical heads on the long pins to prevent link arms against slipping off.				0.985 0.994 0.579
[easy]				
Prolong black lever sticking out to the front by 5cm for better manipulation.				0.985 1.000 0.797
[hard]				
Add third rotor blade to the assembly, same design as the other two, radii on all four long edges, thinner central portion.				0.992 0.936 0.783
[hard]				
Add a connecting hole of 1.7 mm diameter and apply 0.1 mm grooves to increase grip and prevent slipping.				0.987 0.986 0.073